

Vellore Institute of Technology

BCSE301P – Software Engineering Lab

Software Requirements Specification*(Final Consolidated Document)***Learning Continuity and Engagement-Based
E-Learning Platform Management System**

Document Title	Software Requirements Specification – Final Consolidated
Project Name	Learning Continuity and Engagement-Based E-Learning Platform Management System
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive and consolidated description of the Learning Continuity and Engagement-Based E-Learning Platform Management System. The document details the functional and non-functional requirements for the platform, which aims to ensure continuous student participation and genuine learning in online education environments by monitoring engagement, validating attendance through complete video lesson viewing, and enabling timely parental and teacher intervention.

This SRS serves as the foundation for system design and development phases, consolidating all three assessment submissions (Assessment 1 – SRS, Assessment 2 – Project Diagrams, Assessment 3 – Interaction and Architecture Diagrams) into one final reference document.

1.2 Document Conventions

This document follows the standard SRS conventions as defined below:

Term	Description
SHALL	Indicates a mandatory requirement that must be implemented.
SHOULD	Indicates a recommended requirement — highly desirable but not mandatory.
MAY	Indicates an optional requirement considered for future enhancement.
FR-XX	Functional Requirement identifier.
NFR-XX	Non-Functional Requirement identifier.
SR-XX	System Requirement identifier.
IR-XX	Input/Output Requirement identifier.
CR-XX	Communications Requirement identifier.
DR-XX	Data Requirement identifier.

1.3 Intended Audience

This document is intended for the following stakeholders:

Stakeholder	Role
Students	Primary users who access courses, watch videos, and earn attendance.
Teachers	Faculty who upload content, monitor engagement, and initiate interventions.

Stakeholder	Role
Parents	Guardians who receive alerts and monitor student engagement status.
Administrators	System managers who configure, maintain, and oversee the platform.
Development Team	Engineers responsible for designing and implementing the system.
Lab Evaluators	Faculty evaluating the project against BCSE301P requirements.

1.4 Project Scope

The Learning Continuity and Engagement-Based E-Learning Platform Management System is a web-based software solution designed for educational institutions conducting online or blended learning. The system addresses the key shortcomings of traditional LMS platforms — specifically superficial login-based attendance and lack of real student engagement tracking.

In Scope:

- Role-based user authentication and management (Student, Teacher, Parent, Admin).
- Course and video lesson creation and management by teachers.
- Video completion tracking and proof-based attendance validation.
- Engagement score calculation and continuous activity monitoring.
- Automated alert and notification system for teachers and parents.
- Analytics dashboards and downloadable attendance/engagement reports.

Out of Scope:

- Live video conferencing or real-time classroom features.
- Integration with external LMS platforms (e.g., Moodle, Canvas).
- Mobile application development (native iOS or Android apps).
- Payment or fee management modules.

1.5 Definitions, Acronyms and Abbreviations

Term	Description
LMS	Learning Management System
SRS	Software Requirements Specification
E-Learning	Learning conducted through digital platforms and the internet
Engagement Score	Quantitative measure of student participation in learning activities
Attendance Proof	Attendance verified through 100% complete video lesson viewing
ERD	Entity Relationship Diagram
DFD	Data Flow Diagram

Term	Description
STD	State Transition Diagram
UML	Unified Modeling Language
WBS	Work Breakdown Structure
API	Application Programming Interface
VIT	Vellore Institute of Technology

1.6 References

1. IEEE Computer Society. IEEE Standard for Software Requirements Specifications, IEEE Std 830-1998.
2. Sommerville, I. Software Engineering, 10th Edition, Pearson Education.
3. Pressman, R. S. Software Engineering: A Practitioner's Approach, McGraw-Hill.
4. BCSE301P – Software Engineering Lab Guidelines, Vellore Institute of Technology, 2025-2026.

2. Overall Description

2.1 Product Perspective

The Learning Continuity and Engagement-Based E-Learning Platform Management System is a standalone, web-based software system that manages and monitors student participation in online learning environments. It operates as a centralized platform integrating students, teachers, parents, and administrators into a single unified learning management framework.

Unlike traditional e-learning platforms that rely solely on login-based attendance, this system introduces engagement-based mechanisms: video completion tracking, activity monitoring, engagement score calculation, and multi-stakeholder alert systems. The system maintains a centralized relational database for all user data, course content, attendance records, engagement scores, and notification logs.

2.2 Product Functions

The major functions of the system are illustrated in the flowchart below and summarised as follows:

- Role-based user authentication for students, teachers, parents, and administrators.
- Course and video lesson creation and management by teachers.
- Video lesson playback with real-time completion tracking.
- Attendance validation based on 100% video lesson completion.
- Continuous monitoring of student engagement levels and activity logs.
- Identification of inactive or low-engagement students.
- Automated notification system for students, teachers, and parents.
- Generation of attendance and engagement reports for educators.

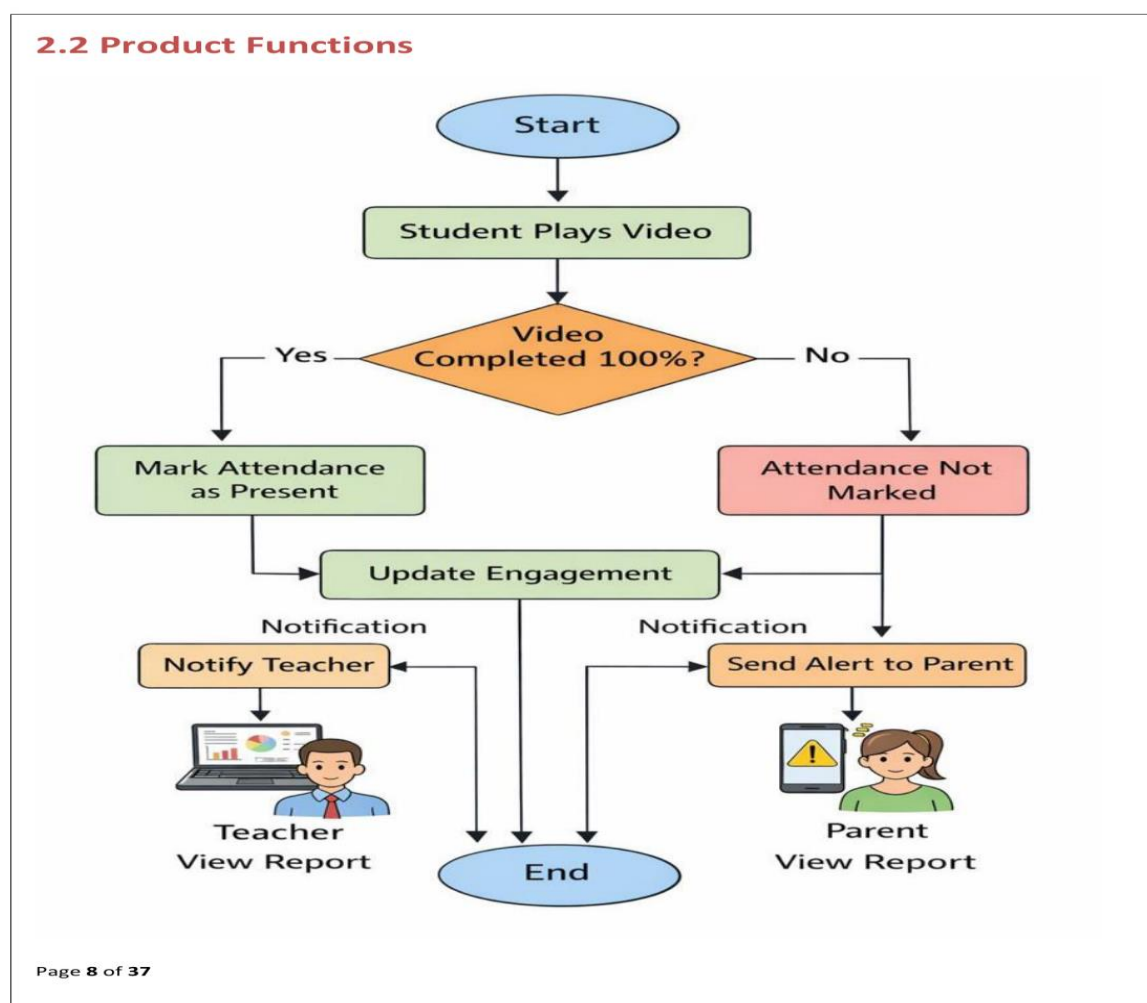


Figure 2.1 – Product Functions Flowchart

2.3 User Classes and Characteristics

User Class	Characteristics and Responsibilities
Student	Accesses enrolled courses and video lessons; watches videos to earn attendance; views personal attendance status and engagement score. Requires basic computer and internet usage skills.
Teacher	Uploads course content and video lessons; monitors student attendance and engagement dashboards; reviews reports and initiates academic intervention. Requires moderate technical familiarity.
Parent	Views student engagement and attendance status; receives automated alerts for low engagement or inactivity. Requires minimal technical knowledge.
Administrator	Manages users, courses, and system settings; maintains overall system functionality; generates system-wide reports. Requires advanced technical knowledge.

2.4 Operating Environment

- Web-based application accessible through standard browsers (Chrome, Firefox, Edge, Safari).
- Server-side application with a centralized relational database.
- Requires stable internet connectivity for video streaming and engagement monitoring.
- Compatible with desktop, laptop, tablet, and mobile devices.

2.5 Design and Implementation Constraints

- Attendance is granted only after 100% video lesson completion — no partial credit permitted.
- Internet access is mandatory for video-based learning and real-time engagement tracking.
- User data privacy and security must be maintained at all times through role-based access control.
- The system must handle multiple concurrent users without performance degradation.

2.6 Assumptions and Dependencies

- Students have access to internet-enabled devices with modern web browsers.
- Teachers regularly upload course content and maintain lesson schedules.
- Parents provide valid contact information (email/phone) during student registration.
- The institution provides server infrastructure and supports online learning operations.
- Third-party email/SMS gateway services are available for notification delivery.

3. Functional Requirements

ID	Requirement
FR-3.1	The system SHALL allow students, teachers, parents, and administrators to log in using role-based authentication with secure credentials.
FR-3.2	The system SHALL support new user registration with profile creation and role assignment by the administrator.
FR-3.3	The system SHALL allow students to access only the courses in which they are enrolled.
FR-3.4	The system SHALL allow teachers to upload video-based learning content in standard formats (MP4, AVI).
FR-3.5	The system SHALL track video playback progress in real time for each student.
FR-3.6	The system SHALL mark attendance only after 100% video lesson completion by the student.
FR-3.7	The system SHALL monitor student learning activities including logins, video views, and session durations.
FR-3.8	The system SHALL calculate and display an engagement score for every student based on activity data.
FR-3.9	The system SHALL identify students with prolonged inactivity or engagement scores below a defined threshold.
FR-3.10	The system SHALL notify teachers when students show low engagement or inactivity beyond a defined threshold.
FR-3.11	The system SHALL notify parents when student engagement falls below a defined threshold or inactivity is detected.
FR-3.12	The system SHALL generate attendance and engagement reports for monitoring and intervention purposes.
FR-3.13	The system SHALL provide role-specific dashboards for students, teachers, parents, and administrators.
FR-3.14	The system SHALL allow students to enroll in available courses and administrators to manage enrollments.
FR-3.15	The system SHALL maintain a complete log of all notifications sent to users for audit purposes.

4. Non-Functional Requirements

ID	Requirement
NFR-4.1	The system SHALL accurately calculate attendance and engagement values without computational error (Accuracy).
NFR-4.2	The system SHALL maintain logs of all attendance records, engagement updates, and notifications sent (Audit Trail).
NFR-4.3	The system SHALL remain available 24/7 with a minimum uptime of 99% during scheduled academic hours (Availability).
NFR-4.4	The system SHALL support a minimum of 500 concurrent users without performance degradation (Capacity).
NFR-4.5	The system SHALL store attendance and engagement data for a minimum of five academic years (Data Retention).
NFR-4.6	The system SHALL support uninterrupted learning and monitoring operations during peak periods (Operations).
NFR-4.7	The system SHALL respond to user actions within 2 seconds under normal operating load (Performance).
NFR-4.8	The system SHALL operate on all major browsers and devices without modification (Portability).
NFR-4.9	The system SHALL recover all data after system failures using automated backup and restoration mechanisms (Recoverability).
NFR-4.10	The system SHALL function consistently without unexpected failures for all core modules (Reliability).
NFR-4.11	The system SHALL protect user data using secure authentication, HTTPS, and role-based access control (Security).
NFR-4.12	The system SHALL send notifications within 5 minutes after disengagement or attendance violations are detected (Timing).
NFR-4.13	The system architecture SHALL support horizontal scaling to accommodate growing user bases (Scalability).
NFR-4.14	The system codebase SHALL follow modular design principles to allow easy maintenance and updates (Maintainability).

5. System Requirements

5.1 Software Requirements

ID	Requirement
SR-5.1	The system SHALL run on standard web browsers (Chrome, Firefox, Edge, Safari).
SR-5.2	The system SHALL use a server-side web application framework (e.g., Node.js / Django / Spring Boot).
SR-5.3	The system SHALL use a relational database management system (e.g., MySQL or PostgreSQL).
SR-5.4	The system SHALL support email or SMS services for automated notification delivery.
SR-5.5	The system SHALL implement HTTPS for all client-server communications.

5.2 Hardware Requirements

ID	Requirement
HR-5.6	The system SHALL require a server with minimum 8-core CPU, 16 GB RAM, and 500 GB SSD storage.
HR-5.7	Users SHALL access the system using desktops, laptops, or mobile devices with modern browsers.
HR-5.8	The system SHALL require stable broadband internet connectivity (minimum 10 Mbps) for video streaming.

5.3 User Requirements

ID	Requirement
UR-5.9	Students SHALL have internet-connected devices to participate in video-based learning.
UR-5.10	Teachers SHALL upload and manage course content on a regular and timely basis.
UR-5.11	Parents SHALL provide valid contact details (email/phone) for receiving automated alerts.

6. Input and Output Requirements

6.1 Input Requirements

ID	Requirement
IR-6.1	The system SHALL accept usernames and passwords for role-based authentication.

ID	Requirement
IR-6.2	The system SHALL accept video lesson files uploaded by teachers in standard formats (MP4, AVI).
IR-6.3	The system SHALL capture learning activity data including video views, login timestamps, and session durations.
IR-6.4	The system SHALL accept parent email addresses and phone numbers for notifications.
IR-6.5	The system SHALL accept course title, description, and associated lesson information from teachers.

6.2 Output Requirements

ID	Requirement
OR-6.6	The system SHALL display real-time attendance status (Present/Absent) for each lesson per student.
OR-6.7	The system SHALL display calculated engagement scores for each student with trend indicators.
OR-6.8	The system SHALL generate and deliver alert notifications to students, teachers, and parents.
OR-6.9	The system SHALL generate downloadable attendance and engagement reports in PDF/Excel format.
OR-6.10	The system SHALL display role-specific dashboards with performance summaries and analytics.

7. Communications Requirements

ID	Requirement
CR-7.1	The system SHALL use secure internet-based client-server communication over HTTPS.
CR-7.2	The system SHALL transmit all sensitive data in encrypted form using TLS.
CR-7.3	The system SHALL deliver notifications through email (SMTP) or SMS gateway services.
CR-7.4	The system SHALL use RESTful APIs for communication between frontend and backend modules.

8. Usability Requirements

ID	Requirement
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ID	Requirement
UR-8.1	The system SHALL provide a simple, intuitive, and user-friendly interface for all user roles.
UR-8.2	The system SHALL support easy navigation with clear menus, breadcrumbs, and labels.
UR-8.3	The system SHALL be fully accessible on both desktop and mobile devices (responsive design).
UR-8.4	The system SHALL require minimal technical training for end users — onboarding within 30 minutes.

9. Data Requirements

ID	Requirement
DR-9.1	The system SHALL store all user, course, attendance, and engagement data in a normalized relational database.
DR-9.2	The system SHALL link engagement data with reporting modules for real-time analytics.
DR-9.3	The system SHALL maintain referential integrity and data consistency across all modules.
DR-9.4	The system SHALL implement automated daily backups with off-site storage capability.

10. External Interface Requirements

ID	Requirement
EIR-10.1	The system SHALL provide access through standard web browsers using HTML5/CSS3/JavaScript frontend.
EIR-10.2	The system SHALL interact with server hardware for processing and database storage.
EIR-10.3	The system SHALL communicate using internet protocols (HTTPS) and third-party messaging services.
EIR-10.4	The system SHALL provide role-specific dashboards for students, teachers, parents, and administrators.

11. Design Constraints

ID	Requirement
DC-11.1	The system SHALL enforce role-based access control — users can only access

ID	Requirement
	features relevant to their role.
DC-11.2	System performance is dependent on server capacity and network bandwidth.
DC-11.3	The system interface SHALL remain simple, consistent, and responsive across all devices.
DC-11.4	The system SHALL use database storage efficiently — video files stored separately from the application database.
DC-11.5	All passwords SHALL be stored in hashed form; no plaintext credentials may be stored in the database.

12. Improvements and Impacts

12.1 Improvements to Existing Capabilities

ID	Requirement
IM-12.1	The system replaces login-based attendance with proof-based, engagement-validated attendance.
IM-12.2	The system enables early detection and intervention for disengaged or at-risk students.
IM-12.3	The system significantly improves parent involvement in the e-learning process through automated alerts.
IM-12.4	The system provides data-driven analytics to help teachers personalise teaching strategies.

12.2 Impacts

Stakeholder	Impact
Students	Improves learning discipline; enables self-monitoring of attendance and engagement.
Teachers	Supports early academic intervention and data-driven course management decisions.
Parents	Increases parental awareness and enables timely involvement in student academic matters.
Institution	Improves overall academic monitoring quality and reduces administrative overhead.

13. Software Process Model – Incremental Model

13.1 Analysis and Justification

The proposed project is an e-learning platform management system that includes multiple functional modules: user authentication, course management, video-based attendance, engagement monitoring, reporting, and notification services. Although the overall system objective is clearly defined, implementing all features in a single cycle would increase complexity and risk.

The Incremental Process Model is selected for the following reasons:

5. The system can be divided into logical, independently deliverable increments.
6. Core functionalities can be developed and validated early, reducing overall project risk.
7. Each increment undergoes requirement analysis, design, coding, and testing, ensuring quality.
8. Changes can be incorporated in later increments without disrupting earlier modules.
9. This model supports systematic, documented development required for academic projects.

13.2 Incremental Development Breakdown

Increment	Objective, Features, and Outcome
Increment 1: Core User Setup	Objective: Establish system foundation. Features: User registration and authentication; role-based access control (Student, Teacher, Parent, Admin); basic dashboards; database initialisation. Outcome: Working login framework with role-specific interfaces.
Increment 2: Course & Content Management	Objective: Enable learning content delivery. Features: Course creation and management by teachers; video lesson upload and storage; course enrollment for students; secure access to learning materials. Outcome: Teachers can upload content; students can access video lessons.
Increment 3: Attendance Validation	Objective: Implement proof-based attendance. Features: Video playback monitoring; detection of 100% video completion; attendance marking only after full completion; attendance record storage. Outcome: Attendance validated on actual learning activity, not login.
Increment 4: Engagement Monitoring	Objective: Measure student participation. Features: Tracking of logins, video views, session durations; engagement score calculation; identification of low-engagement students; teacher dashboards. Outcome: Teachers detect disengagement early through analytics.
Increment 5: Notification System	Objective: Enable timely intervention. Features: Automated alerts for low engagement or inactivity; delivery to teachers and parents; alert log and audit trail. Outcome: Timely corrective actions by parents and teachers.
Increment 6: Reporting and Optimisation	Objective: Finalize and optimise the platform. Features: Attendance and engagement report generation; performance optimisation; data backup and recovery; final testing and validation. Outcome: Stable, fully functional platform ready for deployment.

14. Work Breakdown Structure (WBS)

A Work Breakdown Structure (WBS) is a hierarchical decomposition of the total project work into smaller, manageable components. It helps in effective planning, task allocation, and progress tracking.

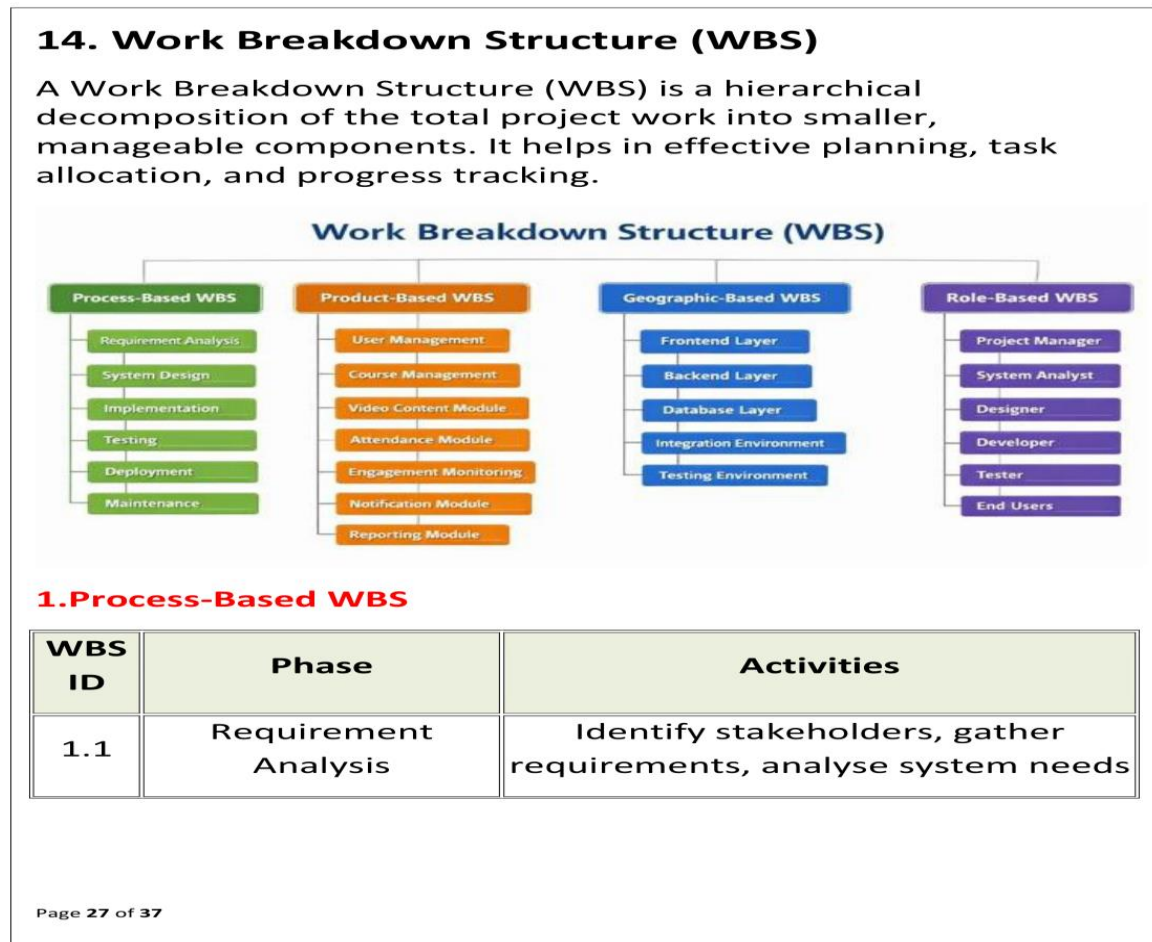


Figure 14.1 – Work Breakdown Structure Overview

14.1 Process-Based WBS

WBS ID / Phase	Activities
1.1 Requirement Analysis	Identify stakeholders, gather requirements, analyse system needs and constraints.
1.2 Requirement Specification	Prepare SRS document, conduct reviews, and validate all requirements.
1.3 System Design	Design system architecture, database schema, API contracts, and user interfaces.
1.4 Module Design	Design attendance, engagement, notification, and reporting modules in detail.

WBS ID / Phase	Activities
1.5 Implementation	Develop frontend (UI) and backend (business logic) modules following the incremental plan.
1.6 Integration	Integrate all developed modules into a unified, cohesive system.
1.7 Testing	Perform unit testing, integration testing, system testing, and user acceptance testing.
1.8 Deployment	Deploy the system to the institutional server environment.
1.9 Maintenance	Monitor performance, fix bugs, apply patches, and implement improvements.

14.2 Product-Based WBS

WBS ID / Component	Description
2.1 User Management Module	Handles user login, authentication, registration, and role management.
2.2 Course Management Module	Manages course creation, enrollment, and lesson structure.
2.3 Video Content Module	Handles video upload, streaming, playback, and access control.
2.4 Attendance Module	Tracks 100% video completion and marks attendance records.
2.5 Engagement Monitoring Module	Calculates engagement scores and detects student inactivity.
2.6 Notification Module	Sends automated alerts to students, teachers, and parents.
2.7 Reporting Module	Generates attendance and engagement reports for all roles.
2.8 Database Layer	Stores all user, course, attendance, engagement, and notification data.

14.3 Geographic-Based WBS

WBS ID / Layer	Responsibilities
3.1 Frontend Layer	Design and development of all user interface components and role-specific dashboards.
3.2 Backend Layer	Implementation of business logic, REST APIs, and system processing rules.
3.3 Database Layer	Data modelling, storage schema design, query optimisation, and retrieval management.
3.4 Integration	Connecting frontend, backend, and database into a unified system.

WBS ID / Layer	Responsibilities
Environment	
3.5 Testing Environment	Validation, debugging, QA testing, and performance benchmarking.

14.4 Role-Based WBS

WBS ID / Role	Responsibilities
4.1 Project Manager	Overall project planning, scheduling, resource allocation, and progress monitoring.
4.2 System Analyst	Requirement analysis, SRS documentation, and system feasibility assessment.
4.3 Designer	User interface design, wireframing, storyboarding, and user experience optimisation.
4.4 Developer	Coding, API development, database integration, and business logic implementation.
4.5 Tester	Test case design, functional testing, regression testing, and defect reporting.
4.6 End Users	System usage, feedback provision, and user acceptance validation.

15. Storyboard

The storyboard outlines the major screens and user interaction flows for each module of the system.

15.1 Login Page

Objective: Allow all users to log into the system securely with role-based credentials.

- Elements: Username field, Password field, 'Forgot Password?' link, Login button.
- Interaction: Users enter valid credentials and click Login. Invalid credentials show an error. Successful login redirects to role-specific dashboard.

15.2 Dashboard

Objective: Display a personalised overview based on user role.

- Elements: Navigation menu (Courses, Attendance, Engagement, Notifications, Reports); widgets for Attendance Status, Engagement Score, and Alerts.
- Students see: learning progress and attendance summary.
- Teachers see: class engagement overview and student alert flags.
- Parents see: child engagement status and notification history.

15.3 Course and Video Management

Objective: Enable teachers to manage courses and students to access learning content.

- Elements: Course list with thumbnails, video lesson player with progress bar, upload button (teachers), enrollment button (students).
- Teachers upload video lessons; students select courses and play lessons.

15.4 Video-Based Attendance System

Objective: Ensure attendance is marked only after complete video lesson viewing.

- Elements: Video player with real-time progress tracker (0%–100%), attendance status indicator (Pending / Completed / Present), skip-prevention controls.
- Attendance is automatically marked only when video reaches 100% completion.

15.5 Engagement Monitoring System

Objective: Track and evaluate student engagement in learning activities.

- Elements: Engagement score display (numerical + graphical), activity log (login history, video views, session durations), trend charts.
- Students view their own engagement score. Teachers monitor all students.

15.6 Teacher Monitoring Dashboard

Objective: Allow teachers to identify disengaged or inactive students and initiate intervention.

- Elements: Engagement analytics table with student-wise breakdown, low-engagement student list, 'Send Notification' action button.

15.7 Parent Alert System

Objective: Notify parents about low engagement or attendance issues for their child.

- Elements: Alert message display, notification history log, student performance summary card.
- Parents receive automated alerts via email/SMS. Parents can log in for detailed status.

15.8 Notification System

Objective: Provide timely and relevant alerts to all stakeholders.

- Elements: Notification bell icon with badge count, alert messages with timestamps, filter by type/date/role.

15.9 Reports and Analytics

Objective: Provide detailed attendance and engagement reports for decision-making.

- Elements: Attendance reports (per student/course/date range), engagement report with score trends, export options (PDF/Excel).
- Teachers view, filter, and download reports. Administrators analyse system-wide metrics.

Appendix A: Analysis Models – Assessment 2 Diagrams

This appendix contains all UML and structural diagrams developed as part of Assessment 2 (17-02-2026) for the Learning Continuity and Engagement-Based E-Learning Platform Management System.

A.1 Entity Relationship Diagram (ERD)

The ER diagram represents the complete data structure of the system. It identifies all major entities, their attributes, and the relationships between them used to track student learning, attendance, and engagement. The system ensures attendance is marked only after video completion and engagement is monitored continuously.

Key Entities: Student, Teacher, Parent, Administrator, Course, Video Lesson, Attendance, Engagement, Notification.

Key Relationships:

- Student enrolls in Course (N:1) — many students can enroll in one course.
- Course contains Video Lessons (1:N) — one course has many video lessons.
- Student views Video Lessons (M:N) — students watch videos from multiple courses.
- Student has Attendance (1:N) — one student has many attendance records.
- Student has Engagement (1:1) — one student has one engagement profile.
- Student receives Notifications (1:N) — one student receives many notifications.
- Teacher teaches Course (1:N) — one teacher teaches many courses.
- Parent monitors Student (1:1) — one parent monitors one student.

Task 1: Entity Relationship Diagram (ERD)

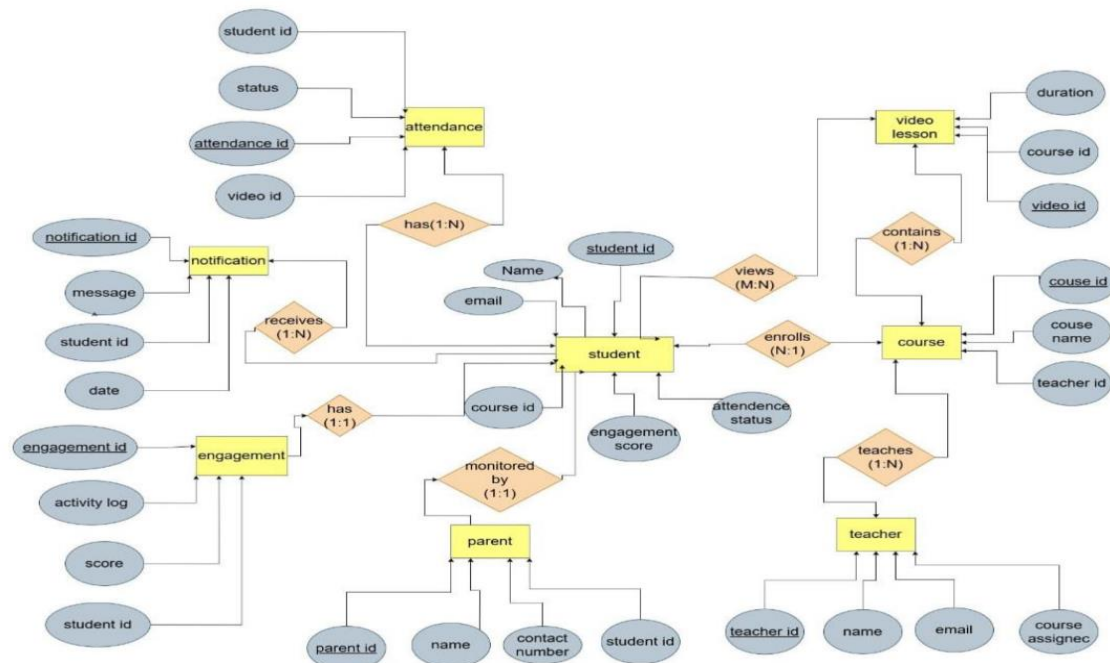


Figure A.1 – Entity Relationship Diagram

A.2 Use Case Diagrams

A.2.1 Learning Use Case Diagram

The Learning Use Case Diagram represents the core learning functionalities. Students can register, log in, enroll in courses, watch video lessons, and resume learning sessions. Teachers create courses, upload video lessons, and manage course content. Administrators manage users and courses.

Task 2: Use Case Diagram

1. Learning Use-Case Diagram

The Learning Use-Case Diagram represents the core learning functionalities of the Learning Continuity and Engagement-Based E-Learning Platform. It shows how students access courses and video lessons, while teachers and administrators manage academic content.

Students can register, log in, enroll in courses, watch video lessons, and resume learning sessions. Teachers create courses, upload video lessons, and manage course content. Administrators manage users and courses to ensure smooth platform operation.

This module ensures structured course delivery and continuous learning access.

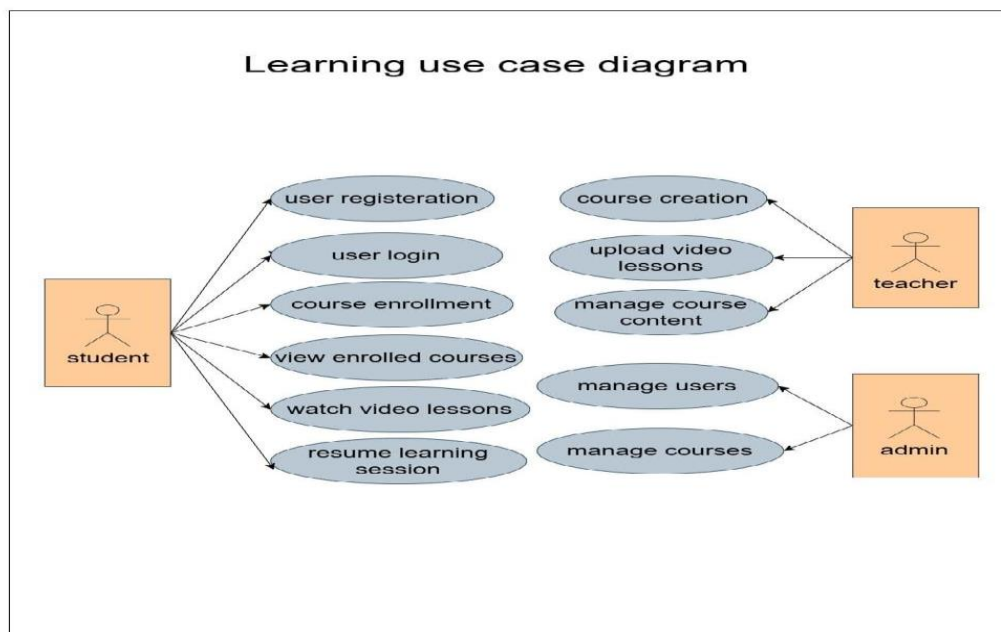


Figure A.2 – Learning Use Case Diagram

A.2.2 Engagement Use Case Diagram

The Engagement Use Case Diagram illustrates how student learning participation is monitored and analysed. The system tracks video completion, logs activity, calculates engagement scores, and identifies low-engagement students. Teachers monitor learning activity and administrators initiate interventions.

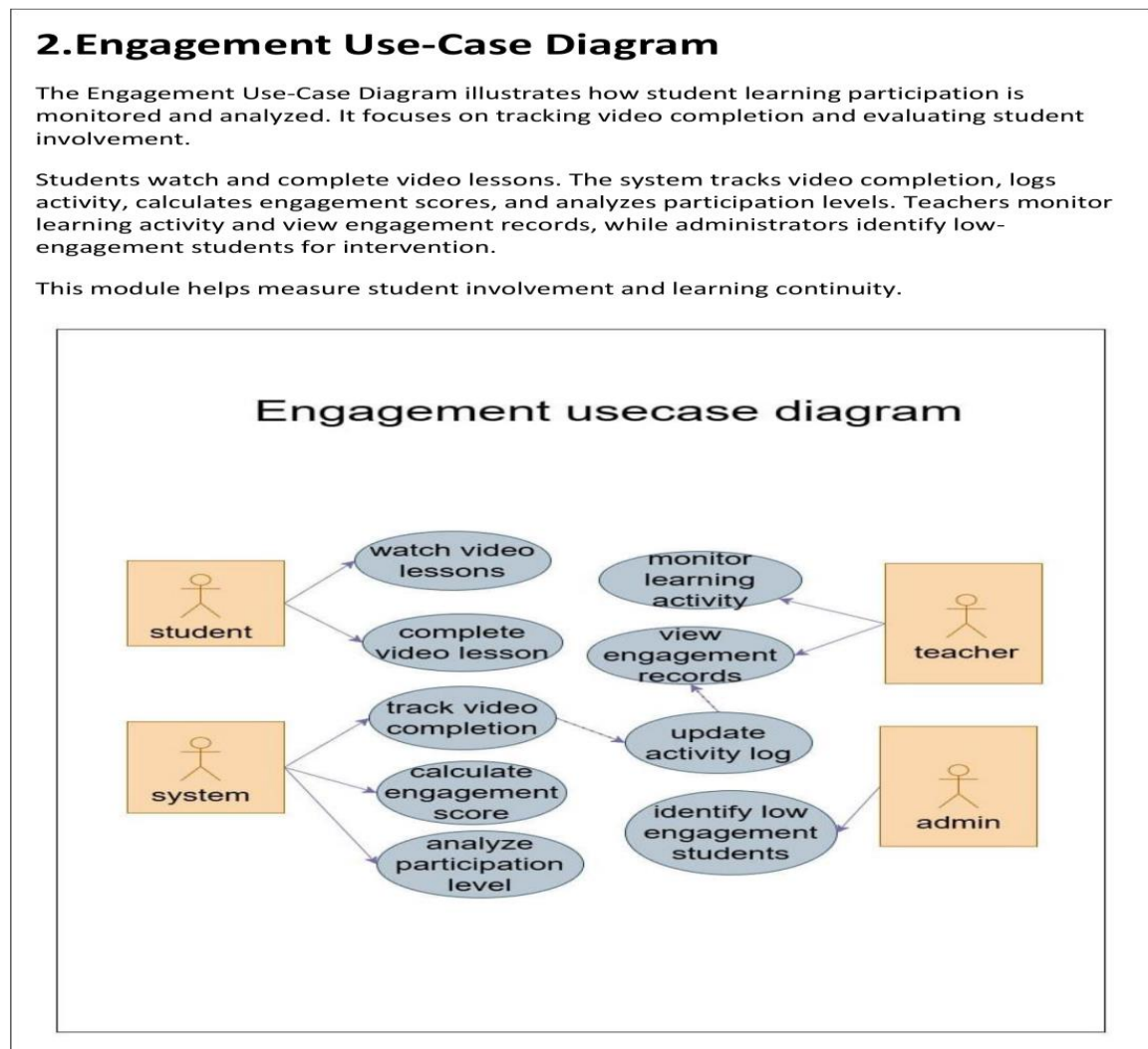


Figure A.3 – Engagement Use Case Diagram

A.2.3 Alert and Notification Management Use Case Diagram

The Alert and Notification Use Case Diagram represents the system's alert generation and communication process. The system detects attendance shortages and low engagement levels. Based on these detections, alerts are generated and notifications are sent to teachers, parents, and students. Teachers can trigger interventions; administrators configure alert thresholds. Teachers can trigger interventions; administrators configure alert thresholds.

3.Alert & Notification Management Use-Case Diagram

The Alert & Notification Use-Case Diagram represents the system's alert generation and communication process.

The system detects attendance shortages and low engagement levels. Based on these detections, alerts are generated and notifications are sent to teachers, parents, and students. Teachers can send warning notifications and trigger interventions, while administrators configure alert thresholds and monitor alerts.

This module ensures timely action to improve student participation and attendance.

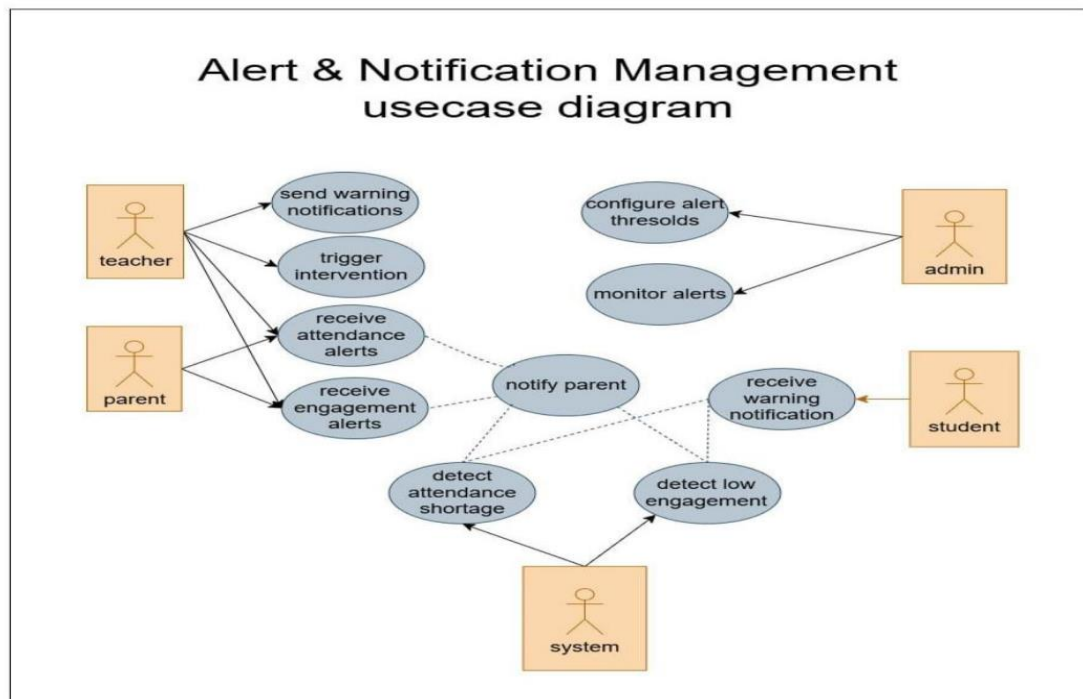


Figure A.4 – Alert and Notification Management Use Case Diagram

A.2.4 Master Use Case Diagram

The Master Use Case Diagram provides a complete functional overview of the entire platform, integrating all stakeholders — students, teachers, parents, and administrators — along with all their respective system interactions across learning, engagement, notification, and reporting modules.

Master Use-Case Diagram

The Master Use-Case Diagram provides an overall view of the entire Learning Continuity and Engagement-Based E-Learning Platform Management System.

It integrates all stakeholders — students, teachers, parents, and administrators — along with their respective system interactions. Students access courses, watch videos, track attendance, and view engagement scores. Teachers manage content and monitor student engagement. Parents view attendance and receive alerts, while administrators manage users, courses, and system operations.

This diagram represents the complete functional scope of the platform.

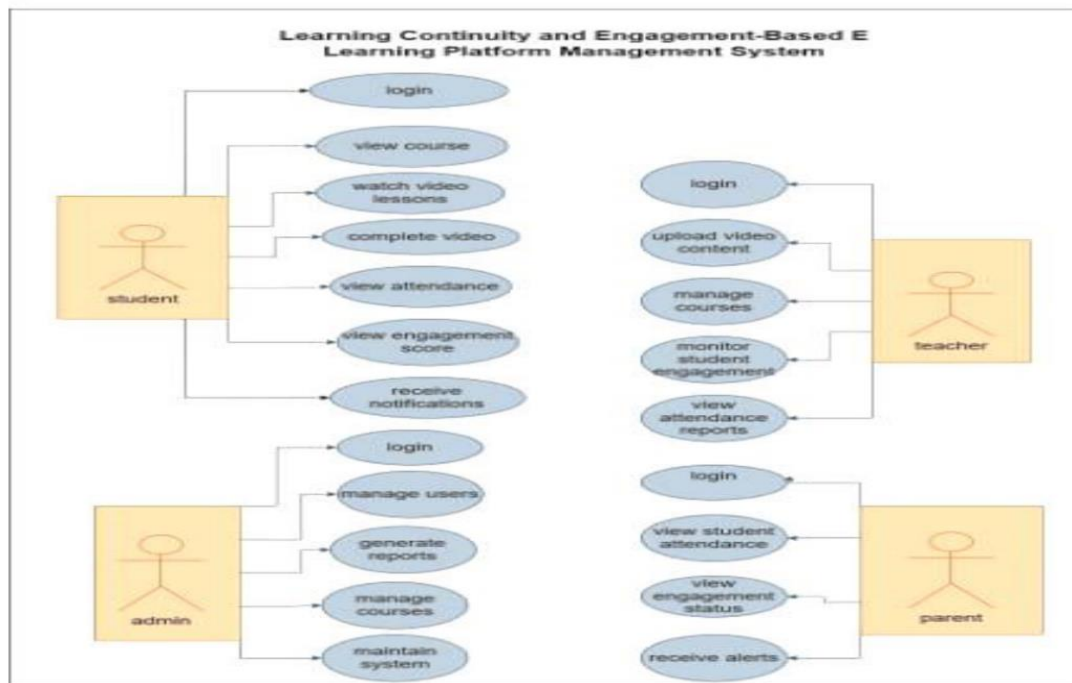


Figure A.5 – Master Use Case Diagram

A.3 State Transition Diagram (STD)

The State Transition Diagram represents the different states of a student in the system throughout their learning journey. The process begins from login where the student registers and enrolls in a course. Once enrolled, the student becomes active by watching video lessons. If the video is completed, attendance is verified, assessment is processed, and a certificate is issued. If inactivity or low engagement is detected, alerts are sent to parents and teachers. The process ends after course completion and session termination.

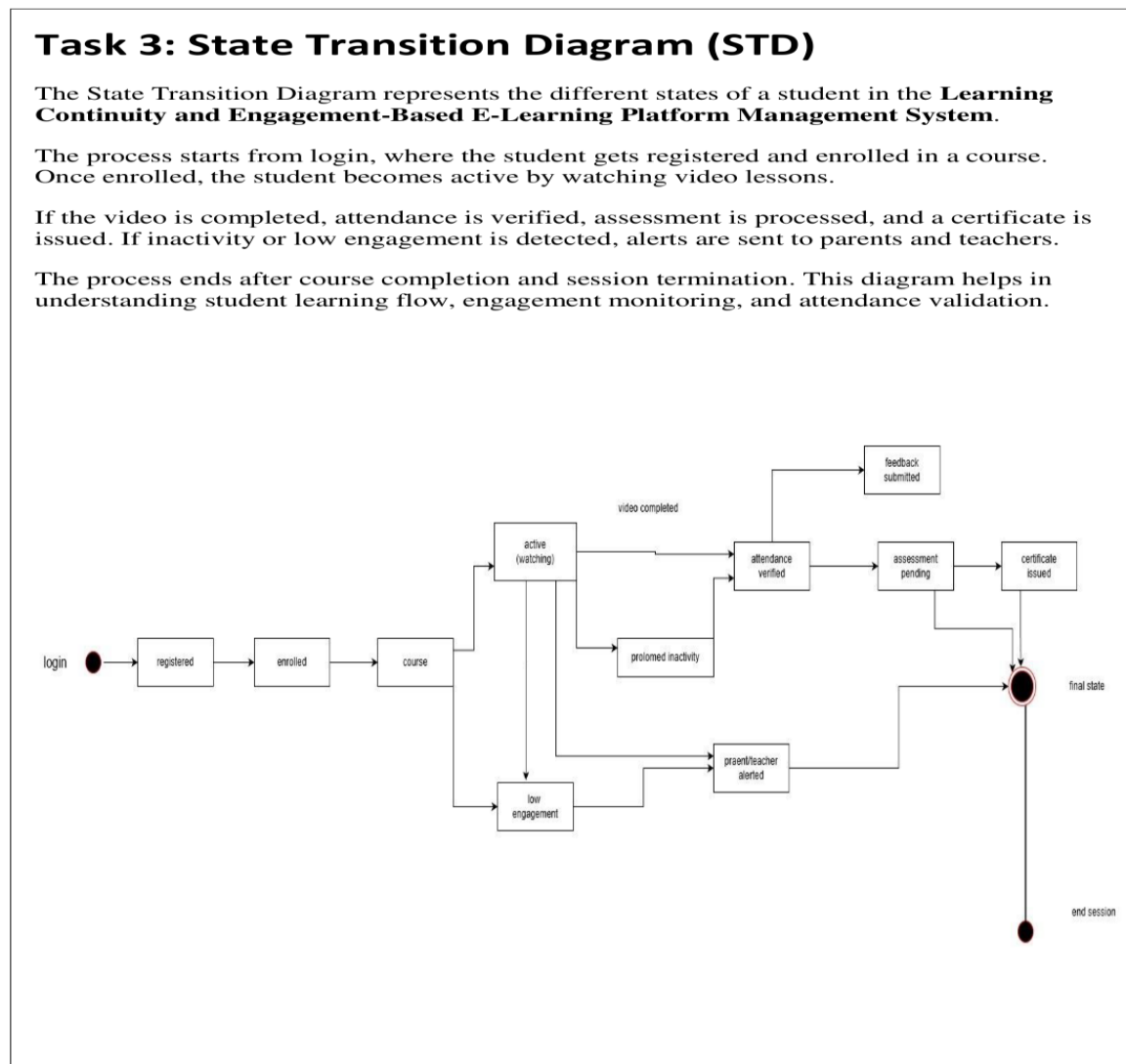


Figure A.6 – State Transition Diagram

A.4 Class Diagram

The class diagram shows the object-oriented structure of the system. The User class is the base class from which Student, Teacher, Parent, and Administrator are derived. The Course class contains VideoLesson objects. The Attendance class validates completion-based attendance. The Engagement class measures participation. The Notification class delivers alerts. The Report class generates analytics outputs.

Task 4: Class Diagram

The class diagram shows the structure of the **Learning Continuity and Engagement-Based E-Learning Platform Management System**. It represents system classes, their attributes, methods, and relationships.

The **User** class is the base class. Student, Teacher, Parent, and Administrator are derived from it. Students access courses and video lessons. Teachers create courses and upload videos. Parents monitor student progress, and admins manage the system.

The **Course** class contains video lessons. The **VideoLesson** class tracks video completion. Attendance is validated using the **Attendance** class. Student participation is measured using the **Engagement** class. Alerts are sent through the **Notification** class, and performance data is generated using the **Report** class.

This diagram helps in understanding the object-oriented design of the system.

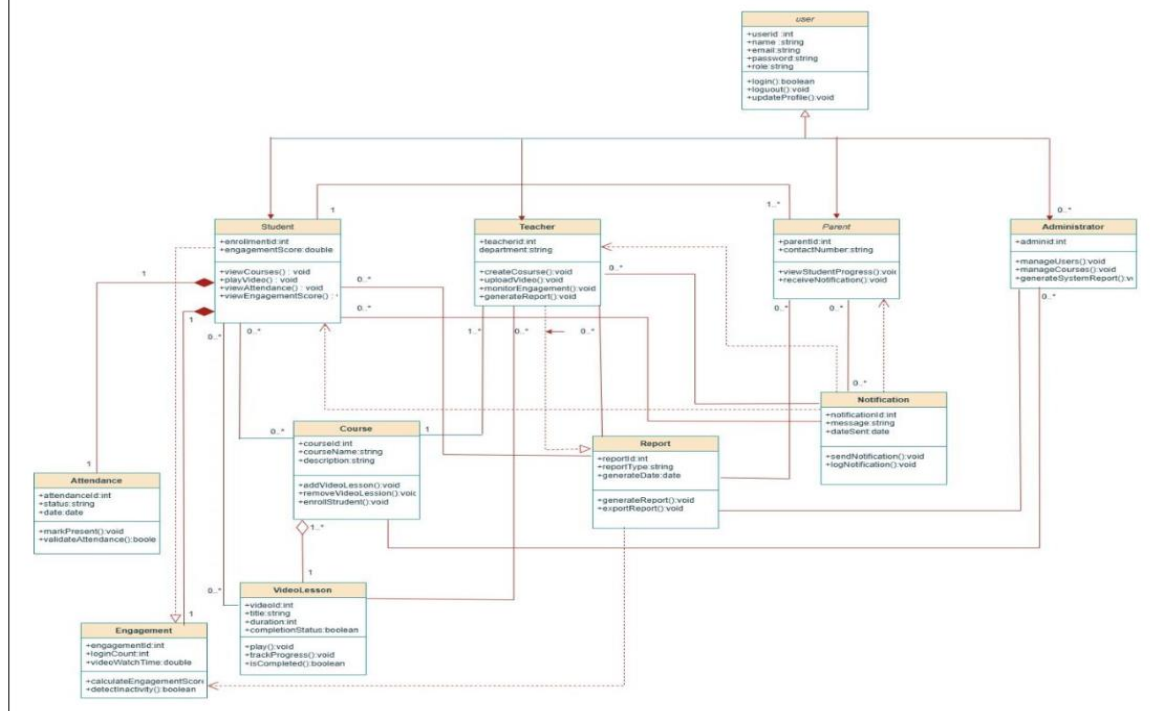


Figure A.7 – Class Diagram

A.5 Data Flow Diagrams (DFD)

A.5.1 Context Level DFD (Level 0)

The Context Level DFD represents the entire system as a single process interacting with four external entities: Student, Teacher, Parent, and Admin. It shows the high-level data flows between stakeholders and the system. Students log in, watch video lessons, and generate learning activity data. Teachers upload content and monitor reports. Admin manages users and courses. The system outputs attendance status, engagement scores, notifications, and alerts.

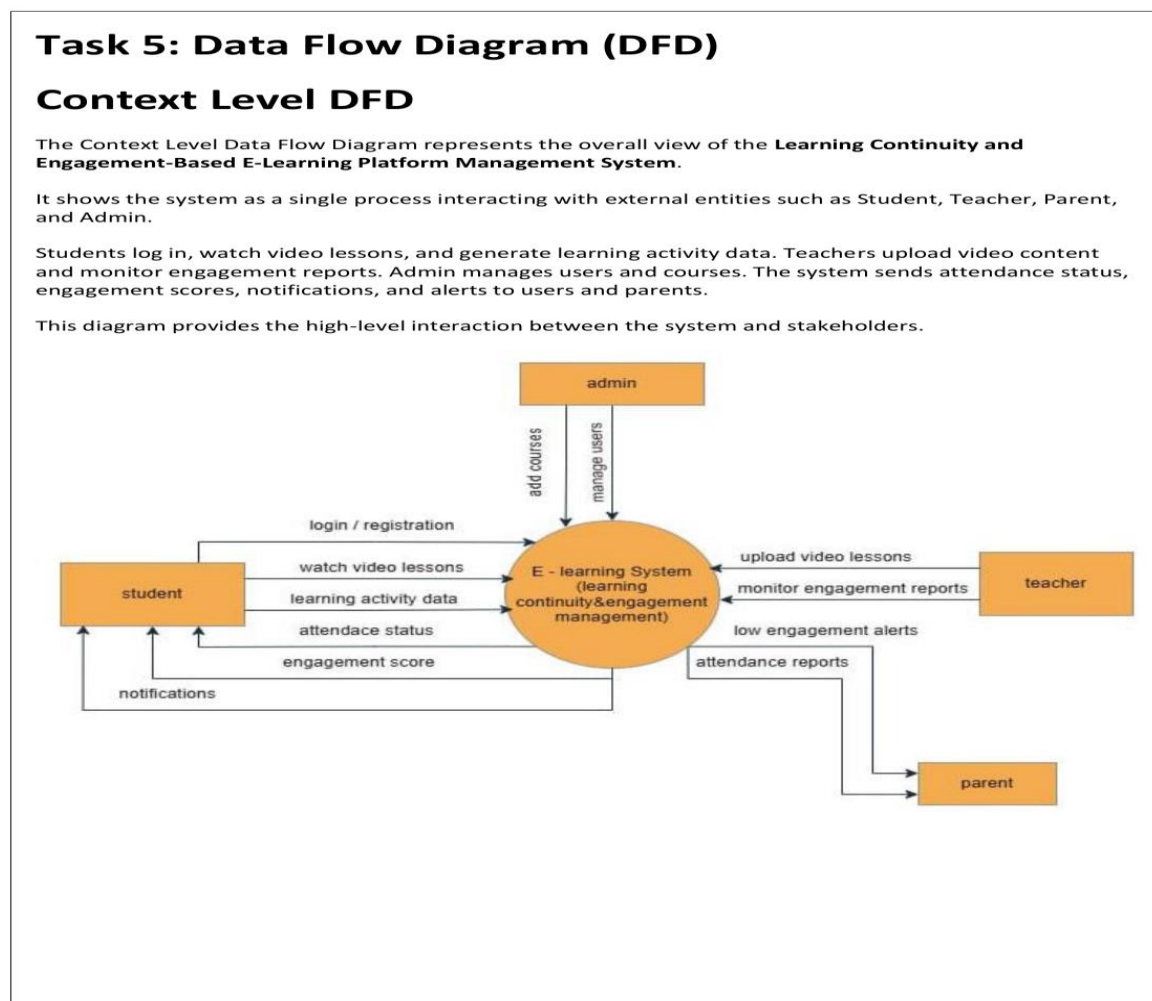


Figure A.8 – Context Level DFD (Level 0)

A.5.2 Level-1 DFD

The Level-1 DFD decomposes the main system into five major functional processes: (1) User Authentication, (2) Course and Video Management, (3) Attendance Validation, (4) Engagement Monitoring, and (5) Notification and Alert System. Students provide login data and access course content. Teachers upload videos and view engagement reports. The attendance module processes video viewing data. Engagement monitoring analyses participation. The notification system delivers targeted alerts.

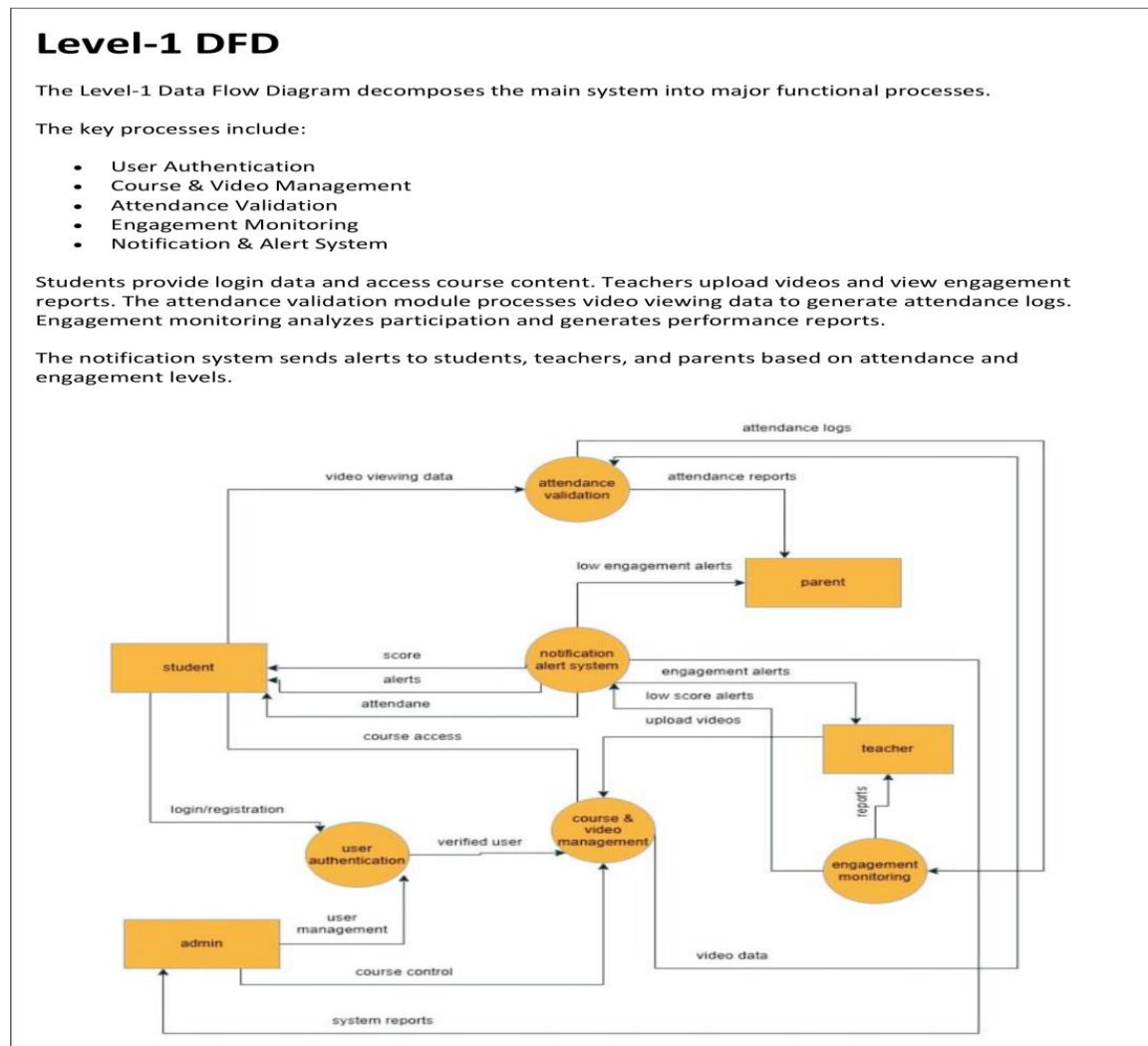


Figure A.9 – Level-1 DFD

A.5.3 Level-2 DFD

The Level-2 DFD provides a detailed breakdown of the Level-1 processes along with data storage interactions. Authentication expands into login validation, role identification, and student registration connected to the User Database. Course and Video Management interacts with the Course and Video databases. Attendance validation stores records in the Attendance Database. Engagement monitoring processes activity logs and stores scores in the Engagement Database. The notification system stores alert history in the Notification Database.

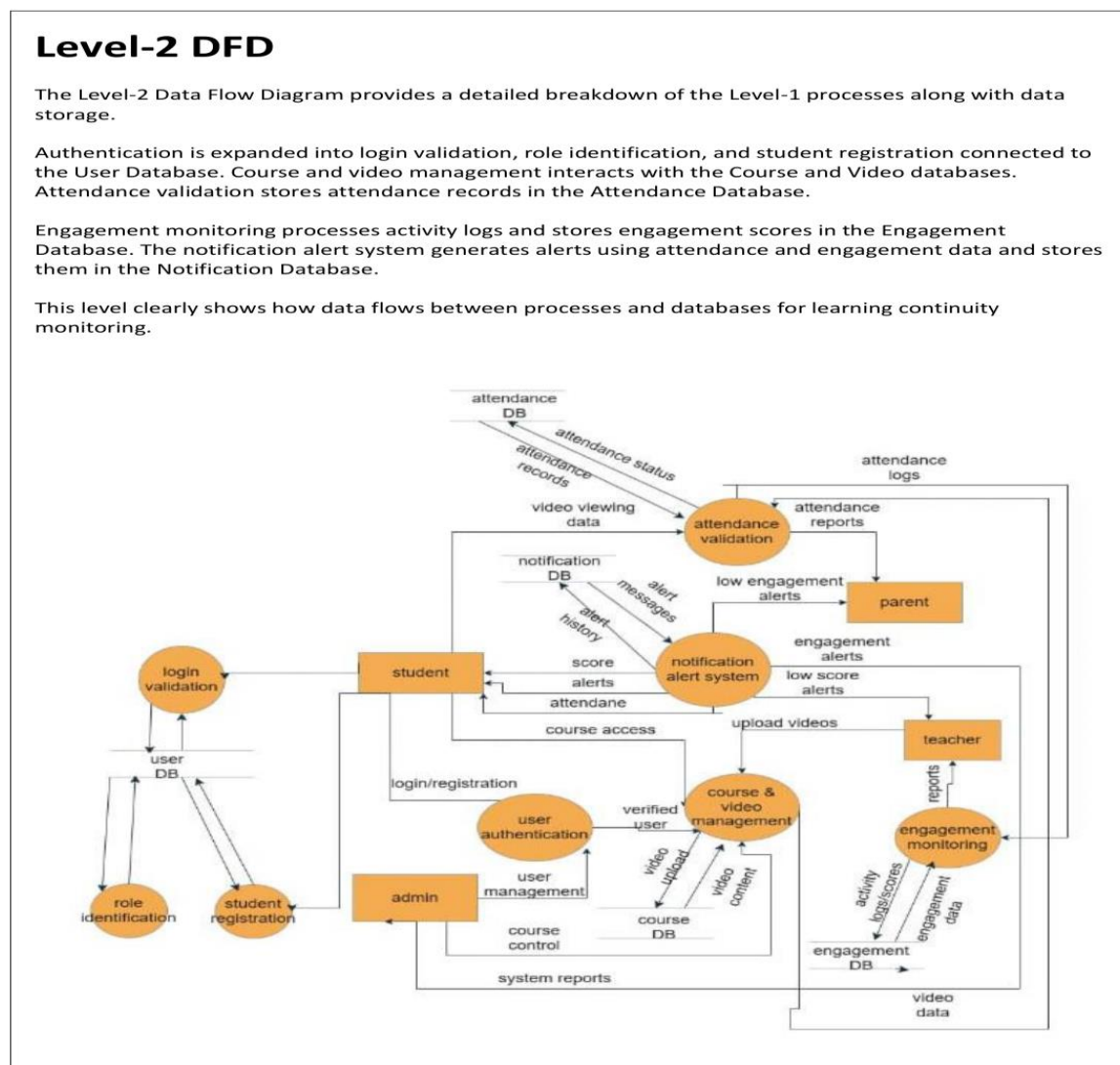


Figure A.10 – Level-2 DFD

Appendix B: Interaction and Architecture Diagrams – Assessment 3

This appendix contains all interaction and architecture diagrams developed as part of Assessment 3 (09-03-2026), providing dynamic interaction and physical deployment perspectives of the system.

B.1 Sequence Diagram (Interaction Model)

The Sequence Diagram models the time-ordered interactions between system objects for the core use case: student login, video lesson access, attendance marking, engagement score update, and conditional alert generation. Key participants include the Student, Parent, E-Learning System, Authentication Service, Course Module, Video Module, Attendance Module, Engagement Module, Notification Module, and four databases (User DB, Course DB, Attendance DB, Engagement DB).

Key Sequence Flow:

10. Student sends Login(username, password) to the E-Learning System.
11. System validates credentials via Authentication Service → fetches user record from User DB.
12. On successful login: Course Module retrieves enrolled courses from Course DB → displays to Student.
13. Student selects a video → Video Module tracks playback progress in real time.
14. ALT [Video Completed]: Attendance Module marks attendance → stores record in Attendance DB.
15. Engagement Module calculates score → stores updated score in Engagement DB.
16. ALT [Low Engagement]: Notification Module sends alert to Parent.
17. ALT [Video Not Completed]: Attendance not marked; engagement updated as partial activity.
18. ALT [Authentication Failed]: System returns invalid credentials message to Student.

Task 1: Sequence Diagram(Interaction Model)

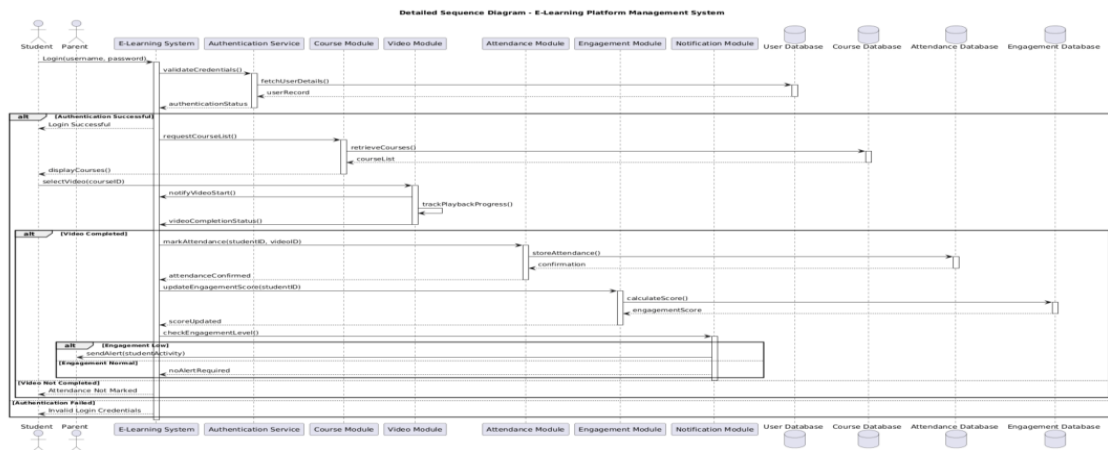


Figure B.1 – Sequence Diagram (Interaction Model)

B.2 Communication Diagram (Interaction Model)

The Communication Diagram shows the same interactions as the Sequence Diagram but organised around structural connections between objects, emphasising communication links over time ordering. Key numbered messages flow from Student → E-Learning System → Authentication Service → Course Module → Video Module → Attendance Module → Engagement Module → Notification Module → Parent.

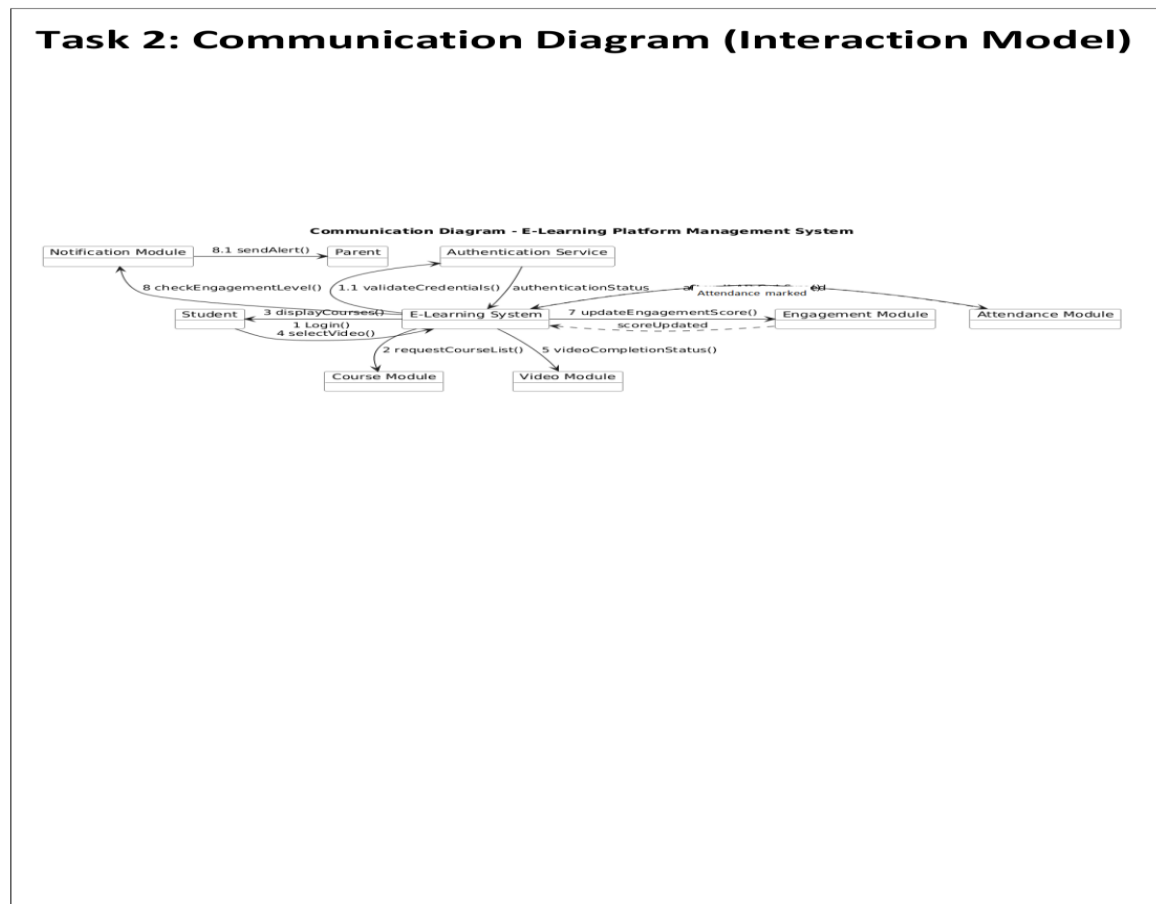


Figure B.2 – Communication Diagram

B.3 Package Diagram

The Package Diagram organises the system's classes into logical packages representing functional layers, forming a vertical dependency hierarchy. Packages include: (1) User Management (UserProfileManager, AuthenticationService, Student, Teacher), (2) Course Management (CourseManager, EnrollmentService, Course), (3) Learning Content (VideoStreamingService, VideoLecture, VideoPlayer), (4) Attendance Management (AttendanceTracker, AttendanceManager), (5) Engagement Monitoring (EngagementAnalyzer, EngagementScore), (6) Notification Services (NotificationService, EmailService), and (7) Data Access Layer (DatabaseConnector, UserDAO, CourseDAO).

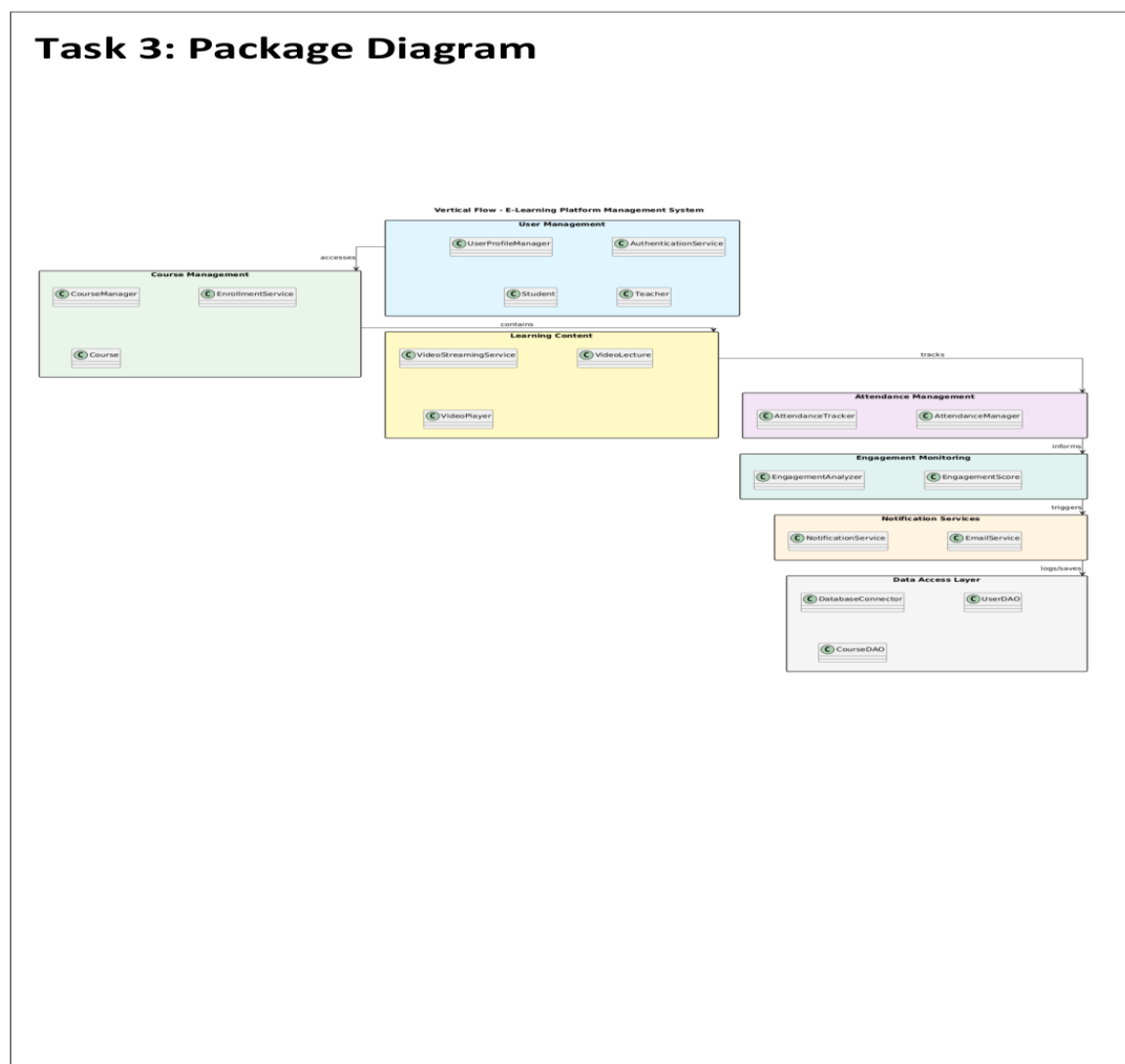


Figure B.3 – Package Diagram

B.4 Component Diagram

The Component Diagram illustrates the physical software components of the system and the interfaces through which they communicate. Key components include Web/Mobile UI, User Management, Authentication Service, Course Management, Video Streaming Service, Attendance Service, Engagement Monitoring Service, Notification Service, and the four databases (Course DB, Attendance DB, Engagement DB, User DB). Video Completion Status flows from Video Streaming Service to Attendance Service, which then informs Engagement Monitoring Service, which triggers Notification Service.

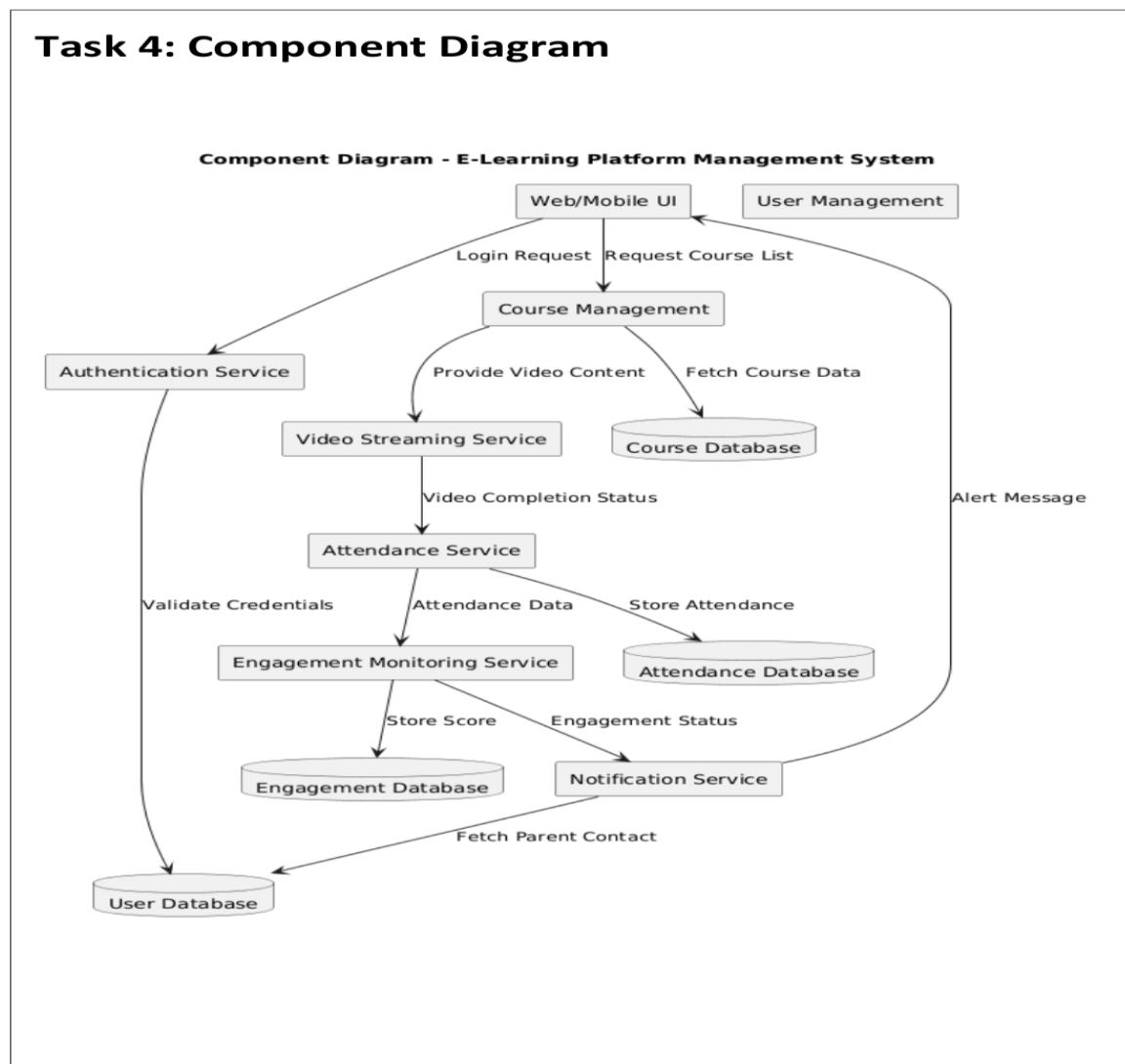


Figure B.4 – Component Diagram

B.5 Deployment Diagram

The Deployment Diagram shows the physical deployment of software components across hardware nodes in the production environment. The architecture comprises: (1) Client Device (Student/Parent/Teacher) running Web Browser or Mobile App communicating via HTTPS; (2) Web Server (Nginx/Apache) with Web Interface Controller and API Gateway; (3) Application Server (Framework) hosting Authentication, User Management, Attendance, Course Management, Video Access, and Engagement Monitoring services; (4) Database Server (MySQL/PostgreSQL) with User DB, Attendance DB, Course DB, and Engagement DB; (5) Video Streaming Server with Video Storage and Streaming Service; (6) Notification Server with Email Notification Service and SMS Notification Service.

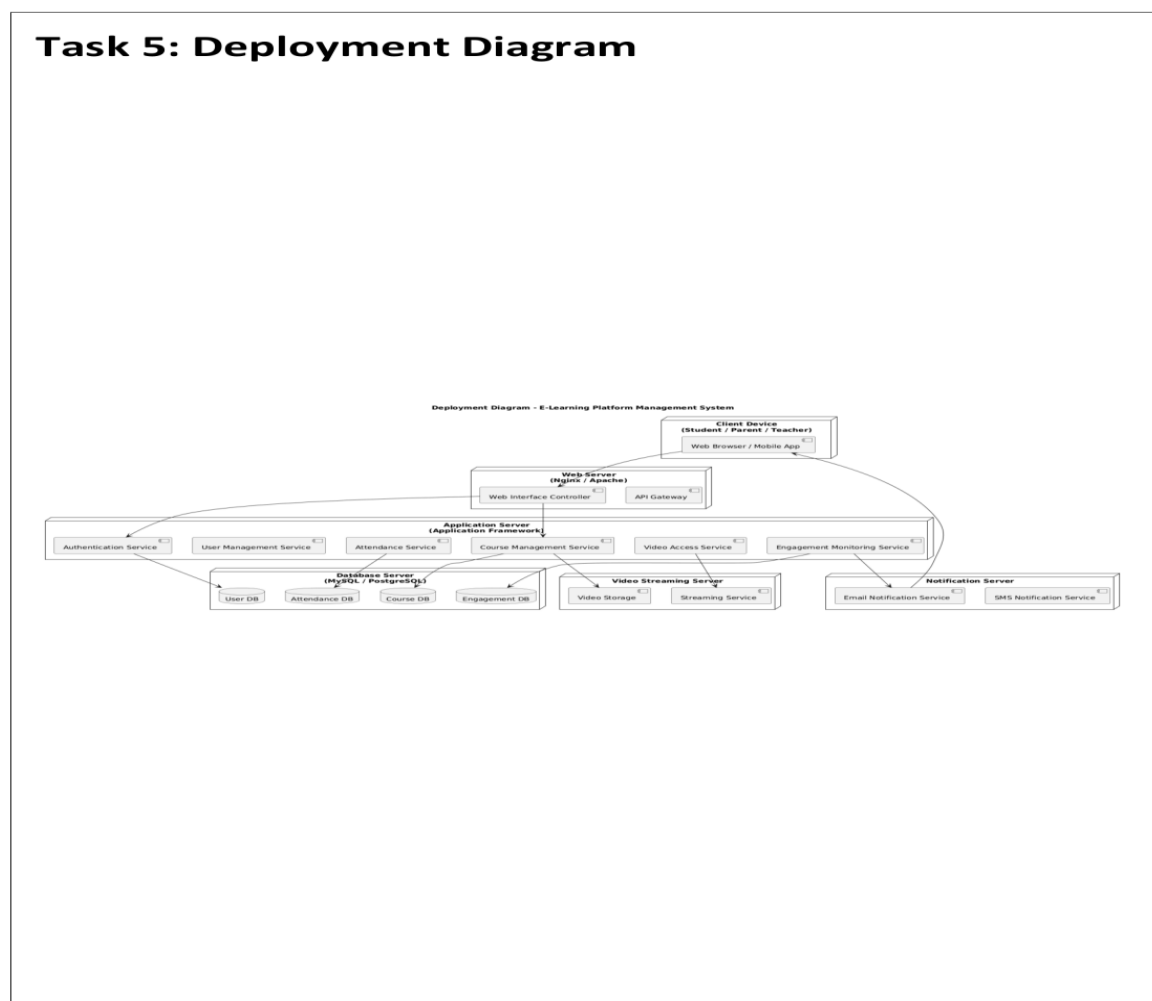


Figure B.5 – Deployment Diagram

Appendix C: Document Version History

Version / Date	Description
v1.0 – 09-01-2026	Initial SRS Document – Assessment 1: Introduction, functional/non-functional requirements, system requirements, process model, WBS, and storyboard.
v1.1 – 17-02-2026	Assessment 2 – Project Diagrams: ERD, Use Case Diagrams (Learning, Engagement, Alert, Master), State Transition Diagram, Class Diagram, and DFD (Context, Level-1, Level-2).
v1.2 – 09-03-2026	Assessment 3 – Interaction and Architecture Diagrams: Sequence Diagram, Communication Diagram, Package Diagram, Component Diagram, and Deployment Diagram.
v2.0 – 08-04-2026	Final Consolidated SRS – All three assessments merged into a single comprehensive document following IEEE 830-1998 SRS format.